

AI2002

Artificial Intelligence

Lecture 17

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Unsupervised Learning

- In **unsupervised learning**, the agent learns patterns in the input even though **no explicit feedback is** supplied.
- **Unsupervised learning** occurs when **no classifications are given and the learner must discover categories and regularities in the data.**
- The most general example of unsupervised learning task is **clustering:**
potentially useful **clusters developed from the input examples.**
- For example, a **taxi agent** might gradually develop a concept of **“good traffic days” and “bad traffic days”.**

Types of Clustering

- There are two types of clustering
 1. K mean Clustering
 2. K Medoid Clustering

K-Mean Clustering

- K-means is a **partitioning clustering** algorithm

- Let the set of data points (or instances) D be

$$\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\},$$

- where
 - $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ir})$ is a **vector** in a real-valued space $X \subseteq R^r$, and
 - r is the number of attributes (dimensions) in the data.
- The k -means algorithm partitions the given data into k clusters.
 - Each cluster **has a cluster center, called centroid.**
 - k is specified by the user

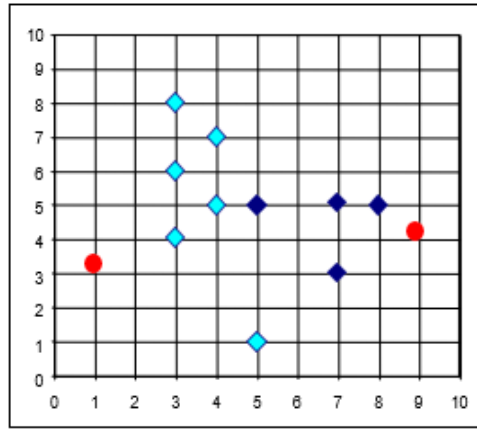
Basic Algorithm

- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

K Mean Clustering

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- ‘Closeness’ is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to ‘Until relatively few points change clusters’

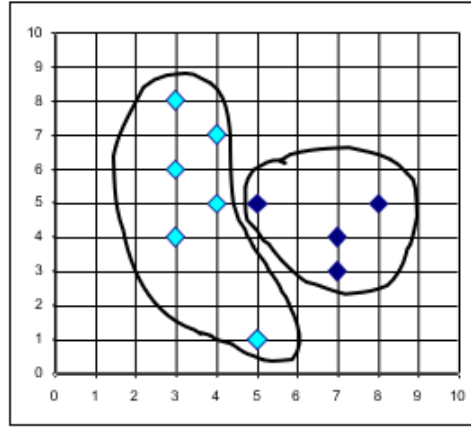
Example



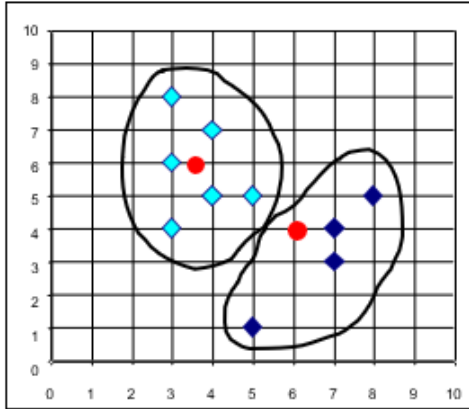
K=2

Arbitrarily choose K
object as initial
cluster center

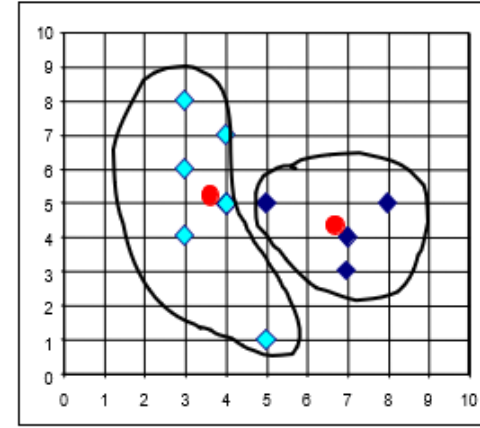
Assign
each
objects
to most
similar
center



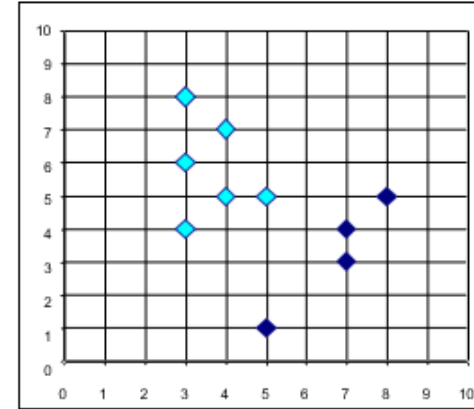
↑ reassign



Update
the
cluster
means



↓ reassign



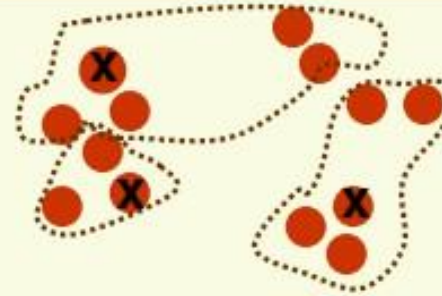
Update
the
cluster
means

K-means Clustering

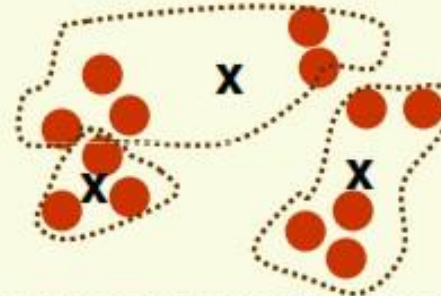
$k = 3$

1. Initialize

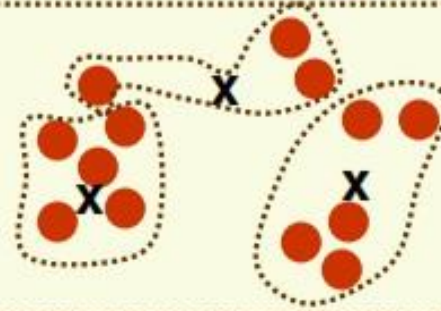
- pick k cluster centers arbitrary
- assign each example to closest center



2. compute sample means for each cluster



3. reassign all samples to the closest mean



4. if clusters changed at step 3, go to step 2

K Mean Clustering Processing

- Pre-processing
 - Normalize the data
 - Eliminate outliers
- Post-processing
 - **Eliminate small clusters** that may represent outliers
 - **Split 'loose' clusters**, i.e., clusters with relatively high SSE
 - **Merge clusters that are 'close'** and that have relatively low SSE

Distance Function

- Most commonly used functions are
 - Euclidean distance and
 - Manhattan (city block) distance
- We denote distance with: $dist(\mathbf{x}_i, \mathbf{x}_j)$, where $\mathbf{x}_i$ and $\mathbf{x}_j$ are data points (vectors)
- They are special cases of Minkowski distance. q is positive integer.

$$d(i, j) = \sqrt[q]{|x_{i1} - x_{j1}|^q + |x_{i2} - x_{j2}|^q + \dots + |x_{ip} - x_{jp}|^q}$$


Example:

- For the given data, compute two clusters using K-means algorithm for clustering where initial cluster centers are (1.0, 1.0) and (5.0, 7.0). Execute for two iterations

Record Number	A	B
R1	1.0	1.0
R2	1.5	2.0
R3	3.0	4.0
R4	5.0	7.0
R5	3.5	5.0
R6	4.5	5.0
R7	3.5	4.5

Solution:

Record Number	Close to C1(1.0, 1.0)	Close to C2(5.0, 7.0)	Assign to cluster
R1(1.0,1.0)	dist(R1, C1)=0.0	dist(R1, C2)=7.21	Cluster1
R2(1.5,2.0)	dist(R2, C1)=1.12	dist(R2, C2)=6.12	Cluster1
R3(3.0,4.0)	dist(R3, C1)=3.61	dist(R3, C2)=3.61	Cluster1
R4(5.0,7.0)	dist(R4, C1)=7.21	dist(R4, C2)=0.0	Cluster2
R5(3.5,5.0)	dist(R5, C1)=4.12	dist(R5, C2)=2.5	Cluster2
R6(4.5,5.0)	dist(R6, C1)= 5.31	dist(R6, C2)=2.06	Cluster2
R7(3.5,4.5)	dist(R7,C1)=4.30	dist(R7, C2)=2.92	Cluster2

Cluster1 {R1, R2} and Cluster2 {R3, R4, R5, R6, R7}.

Calculate the SSE or WCSS

- sum of squared error (SSE) or within cluster sum of squared error (WCSS) is calculated by
- Point (5,2)
- Centroid(3, 3)
- SSE

$$(3 - 5)^2 + (3 - 2)^2$$
$$4 + 1 = 5$$

Centroid Updating

- $C1 = (1.0+1.5)/2, (1.0+2.0)/2$
 - $= 2.50/2, 3.0/2$
 - $= 1.25, 1.5$
- $C2 = (3.0+5.0+3.5+4.5+3.5)/5, (4+7+5+5+4.5)/5$
 - $= 19.5/5, 25.5/5$
 - $= 3.9, 5.1$

Iteration 2

Record Number	Close to C1(1.83, 2.33)	Close to C2(4.12, 5.37)	Assign to cluster
R1(1.0,1.0)	dist(R1, C1)=1.57	dist(R1, C2)=5.37	Cluster1
R2(1.5,2.0)	dist(R2, C1)=0.47	dist(R2, C2)=4.27	Cluster1
R3(3.0,4.0)	dist(R3, C1)=2.04	dist(R3, C2)=1.77	Cluster2
R4(5.0,7.0)	dist(R4, C1)=5.64	dist(R4, C2)=1.85	Cluster2
R5(3.5,5.0)	dist(R5, C1)=3.15	dist(R5, C2)=0.72	Cluster2
R6(4.5,5.0)	dist(R6, C1)=3.78	dist(R6, C2)=0.53	Cluster2
R7(3.5,4.5)	dist(R7,C1)=2.74	dist(R7, C2)=1.07	Cluster2

K Means Clustering

- Time complexity for K-means clustering is

$$O(n \times K \times I \times d)$$

n = number of points,
 K = number of clusters,
 I = number of iterations,
 d = number of attributes

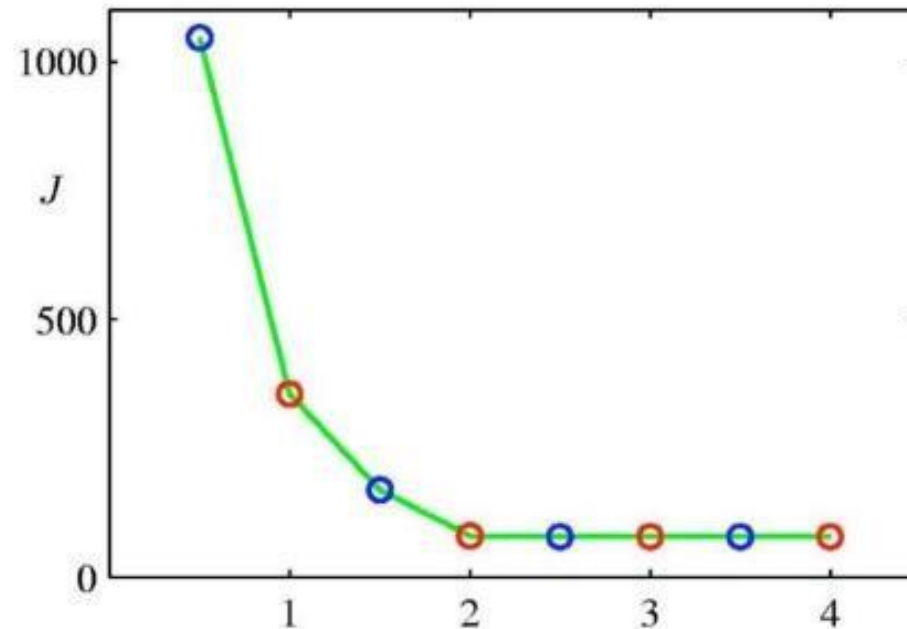
- The storage required is

$$O((n + K)d)$$

n = number of points,
 K = number of clusters,
 d = number of attributes

The Value of K

- One way to select K for the K -means algorithm is to try different values of K , plot the K -means objective versus K , and look at the "elbow-point" in the plot



- For the above plot, $K = 2$ is the elbow point

Elbow Method

WCSS measures **how well the data points are clustered around their respective centroids**. It is defined as the **sum of the squared distances between each point and its cluster centroid**:

$$\text{WCSS} = \sum_{i=1}^k \sum_{j=1}^{n_i} \text{distance}(x_j^{(i)}, c_i)^2$$

$\text{distance}(x_j^{(i)}, c_i)$ represents the distance between the j -th data point $x_j^{(i)}$ in cluster i and the centroid c_i of that cluster.

Example:

Record Number	A	B
R1	1.0	1.0
R2	1.5	2.0
R3	3.0	4.0
R4	5.0	7.0
R5	3.5	5.0
R6	4.5	5.0
R7	3.5	4.5

Use Elbow Method to find the Number of Clusters

K = 1

$$\frac{1.0+1.5+3.0+5.0+3.5+4.5+3.5}{7}, \frac{1.0+2.0+4.0+7.0+5.0+5.0+4.5}{7}$$

{3.14, 4.07}

WCSS

Point (1.0, 1.0)

$$(1.0 - 3.14)^2 + (1.0 - 4.07)^2 = (-2.14)^2 + (-3.07)^2 = 4.58 + 9.42 = 14.00$$

Point (1.5, 2.0)

$$(1.5 - 3.14)^2 + (2.0 - 4.07)^2 = (-1.64)^2 + (-2.07)^2 = 2.69 + 4.29 = 6.98$$

Point (3.0, 4.0)

$$(3.0 - 3.14)^2 + (4.0 - 4.07)^2 = (-0.14)^2 + (-0.07)^2 = 0.02 + 0.005 = 0.025$$

Point (5.0, 7.0)

$$(5.0 - 3.14)^2 + (7.0 - 4.07)^2 = (1.86)^2 + (2.93)^2 = 3.46 + 8.58 = 12.04$$

Point (3.5, 5.0)

$$(3.5 - 3.14)^2 + (5.0 - 4.07)^2 = (0.36)^2 + (0.93)^2 = 0.13 + 0.86 = 0.99$$

Point (4.5, 5.0)

$$(4.5 - 3.14)^2 + (5.0 - 4.07)^2 = (1.36)^2 + (0.93)^2 = 1.85 + 0.86 = 2.71$$

Point (3.5, 4.5)

$$(3.5 - 3.14)^2 + (4.5 - 4.07)^2 = (0.36)^2 + (0.43)^2 = 0.13 + 0.18 = 0.31$$

K Value

- The K value for $K = 1$ is 37.06.

Elbow Method

K	WCSS
1	37.06
2	34.5
3	19.8
4	15.2
5	12.4
6	10.8
7	0

Limitation in K mean Clustering

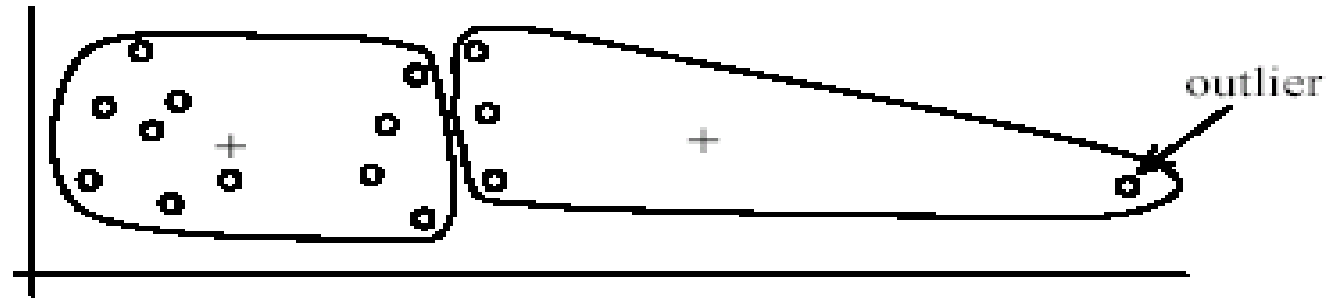
- K-means has problems when the data contains outliers

*The K-means algorithm is very sensitive to the **initial seeds**.*

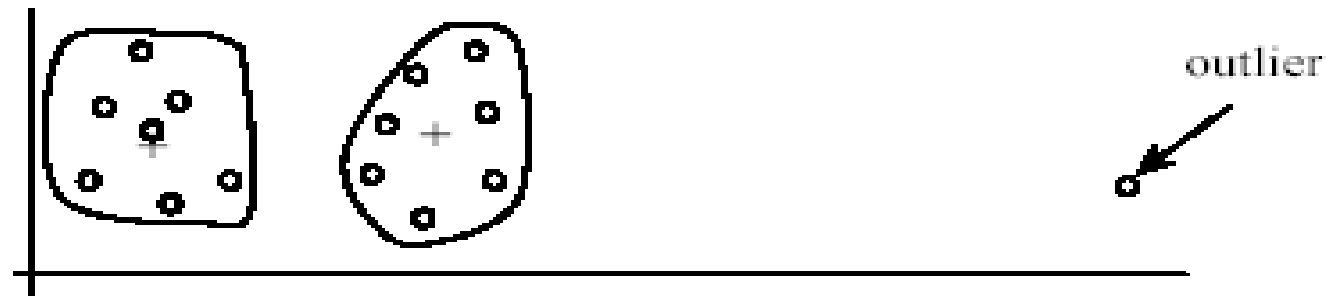
- K-means has problems when clusters are of different
Sizes
Densities
Non-globular shapes

Limitation in K mean Clustering

- K Means have problem when data contain outliers



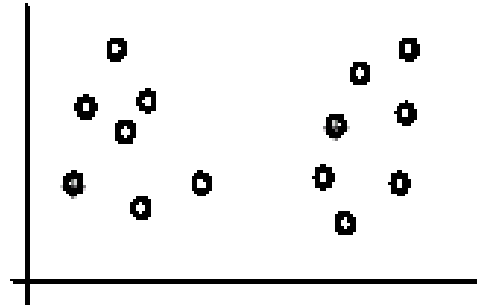
(A): Undesirable clusters



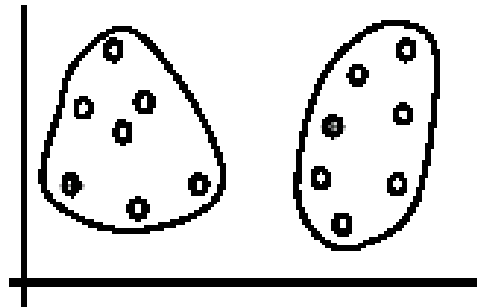
(B): Ideal clusters

Limitation in K mean Clustering

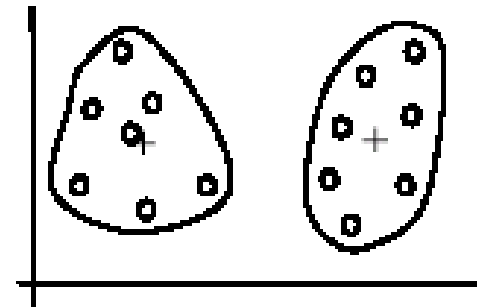
- The algorithm is sensitive to the initial seeds or randomly choose seeds.



(A). Random selection of k seeds (centroids)



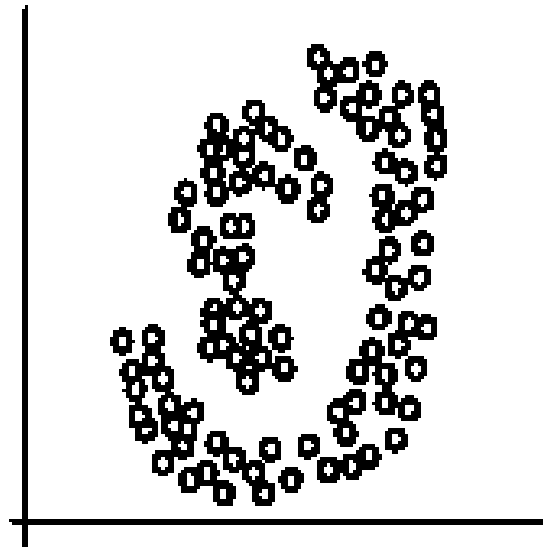
(B). Iteration 1



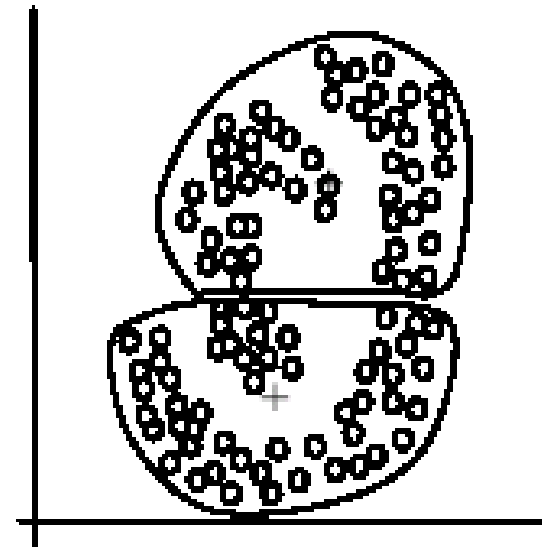
(C). Iteration 2

Limitation in K mean Clustering

- The k -means algorithm is not suitable for discovering clusters that are not hyper-ellipsoids (or hyper- spheres).



(A): Two natural clusters



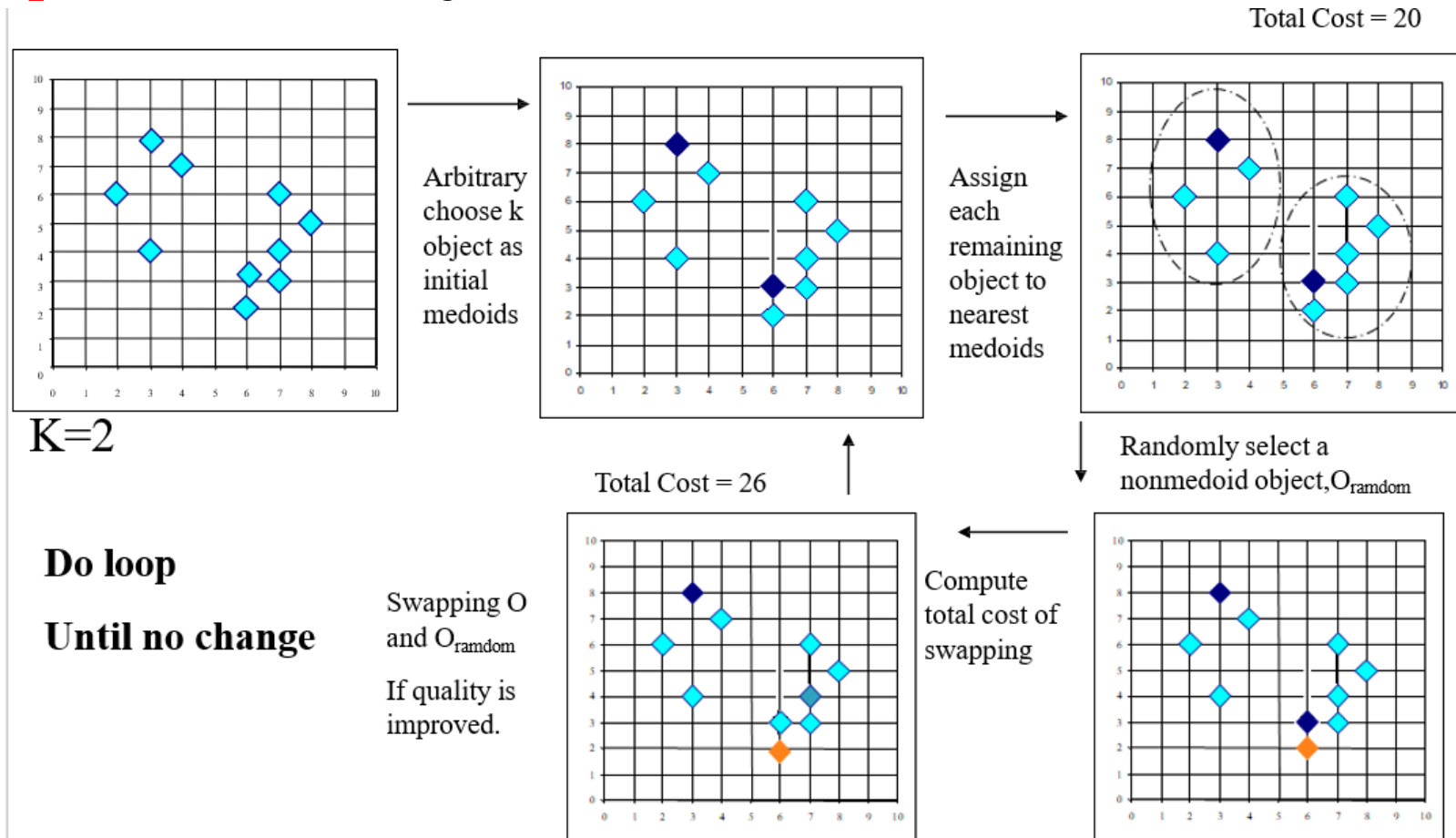
(B): k -means clusters

K Medoid Clustering

- The k-means algorithm is sensitive to outliers!
Since an object with an extremely large value may substantially distort the distribution of the data.
- K-Medoids:
- Instead of taking the mean value of the object in a cluster as a reference point, *medoids can be used*, which is the most centrally located object in a cluster.

K Medoids Clustering

- Find *representative* objects, called medoids, in the clusters



K Medoid Clustering

- Use real object to represent the cluster

Select k representative objects arbitrarily

For each pair of non-selected object h and selected object

i , calculate the total swapping cost TC_{ih}

For each pair of i and h ,

If $TC_{ih} < 0$, i is replaced by h

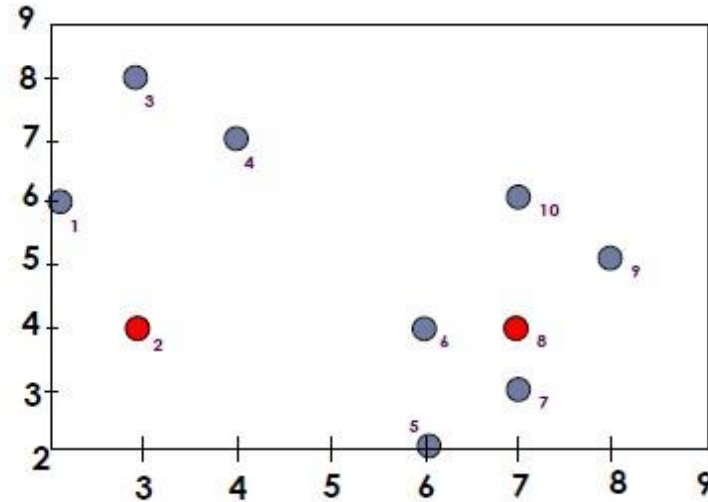
Then assign each non-selected object to the most similar representative object

repeat steps 2-3 until there is no change

Example

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



Goal: create two clusters

Choose randomly two medoids

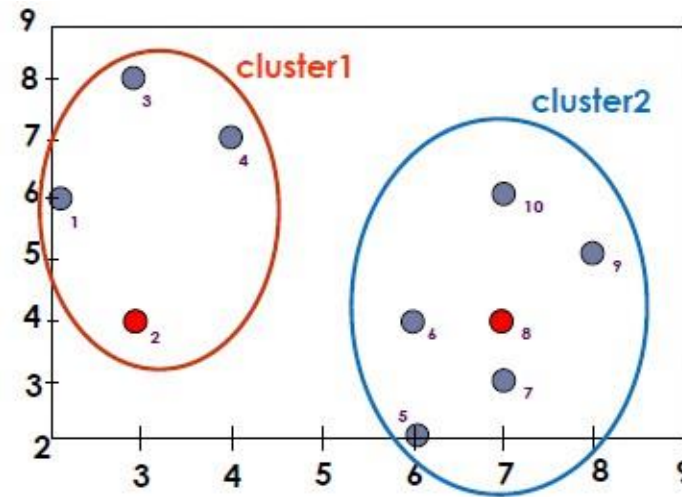
$$O_2 = (3, 4)$$

$$O_8 = (7, 4)$$

Solution:

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



→ Assign each object to the closest representative object

→ Using L1 Metric (Manhattan), we form the following clusters

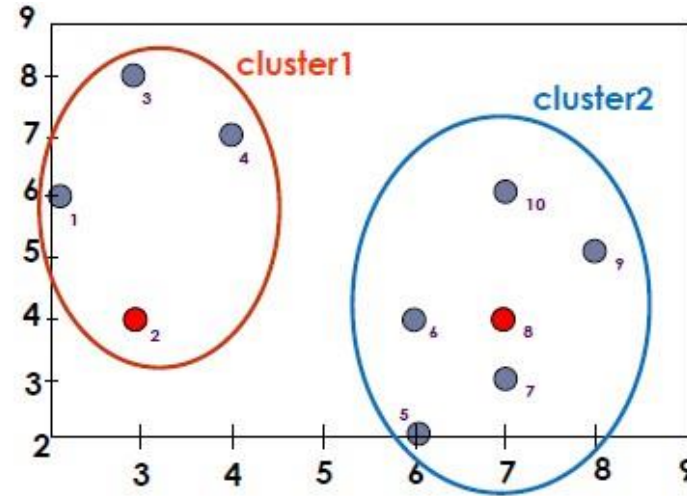
$$\text{Cluster1} = \{O_1, O_2, O_3, O_4\}$$

$$\text{Cluster2} = \{O_5, O_6, O_7, O_8, O_9, O_{10}\}$$

Solution:

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



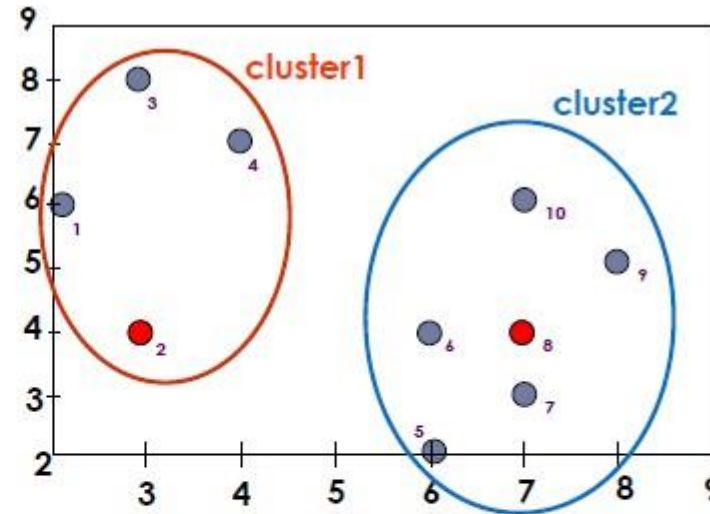
→ Compute the absolute error criterion [for the set of Medoids (O2, O8)]

$$E = \sum_{i=1}^k \sum_{p \in C_i} |p - o_i| = |o_1 - o_2| + |o_3 - o_2| + |o_4 - o_2| + |o_5 - o_8| + |o_6 - o_8| + |o_7 - o_8| + |o_9 - o_8| + |o_{10} - o_8|$$

Solution:

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



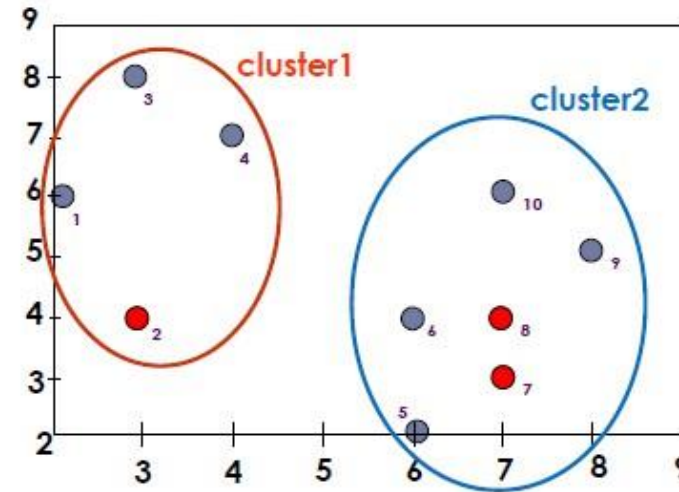
→ The absolute error criterion [for the set of Medoids (O2, O8)]

$$E = (3 + 4 + 4) + (3 + 1 + 1 + 2 + 2) = 20$$

Solution:

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



→ Choose a random object O_7

→ Swap O_8 and O_7

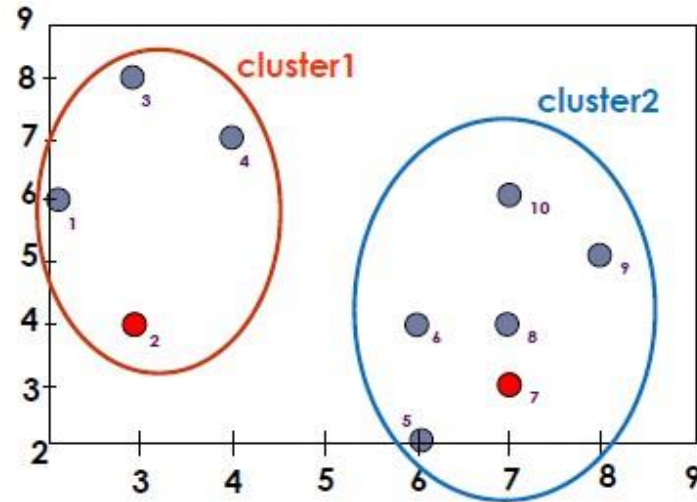
→ Compute the absolute error criterion [for the set of Medoids (O_2, O_7)]

$$E = (3 + 4 + 4) + (2 + 2 + 1 + 3 + 3) = 22$$

Solution:

Data Objects

	A_1	A_2
O_1	2	6
O_2	3	4
O_3	3	8
O_4	4	7
O_5	6	2
O_6	6	4
O_7	7	3
O_8	7	4
O_9	8	5
O_{10}	7	6



→ Compute the cost function

Absolute error [for O_2, O_7] – Absolute error [O_2, O_8]

$$S = 22 - 20$$

$S > 0 \Rightarrow$ it is a bad idea to replace O_8 by O_7

K Medoid Clustering

- *PAM is more robust than k-means in the presence of noise and outliers* because a medoid is less influenced by outliers or other extreme values than a mean
- PAM works efficiently for small data sets but **does not scale well** for large data sets.
- $O(k(n - k)^2)$ for each iteration
where n is # of data points,
 k is # of clusters